A Spatial Analysis of PFAS Pollution in Michigan

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Introduction

- Per- and polyfluoroalkyl substances (PFAS) are synthetic "forever" chemicals that are widely used in everyday products.[1]
- PFAS do not break down easily in the environment or the human body.
- Research indicates that PFAS exposure may be linked to harmful health effects, so monitoring PFAS pollution levels is important for public health research and evidence-based policy decisions.
- We analyzed data on PFAS pollution from the Michigan Department of Environmental Great Lakes and Energy and the Michigan PFAS Action Response Team[2], summarizing patterns in PFAS pollution in Michigan.

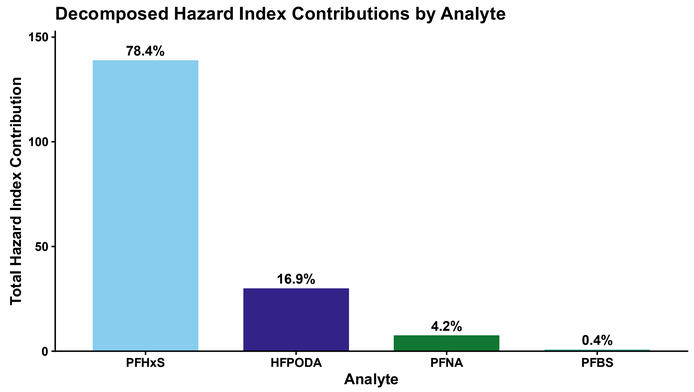
Hazard Index

- The hazard index is a unitless metric from the Environmental Protection Agency (EPA) that summarizes the adverse health effects associated with exposure to mixtures of different PFAS subtypes called analytes.[3]

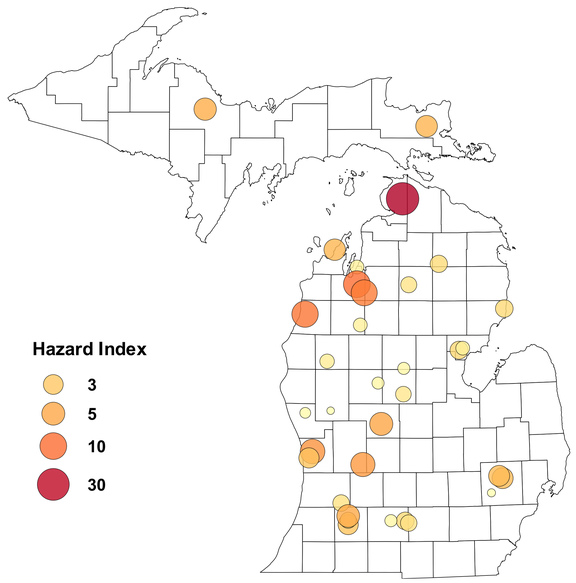
$$\text{Hazard Index} = \left(\frac{\text{HFPODA ppt}}{10\text{ ppt}}\right) + \left(\frac{\text{PFBS ppt}}{2000\text{ ppt}}\right) + \left(\frac{\text{PFNA ppt}}{10\text{ ppt}}\right) + \left(\frac{\text{PFHxS ppt}}{10\text{ ppt}}\right)$$

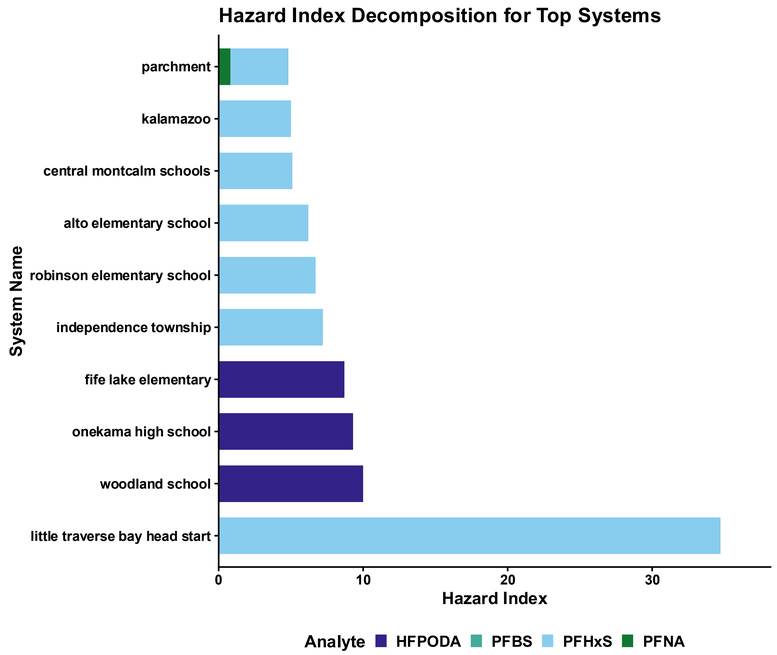
- The EPA's Hazard Index Maximum Contaminant Level (MCL) is set at 1, meaning there are potential elevated health risks for drinking water with hazard index levels exceeding 1.

Results

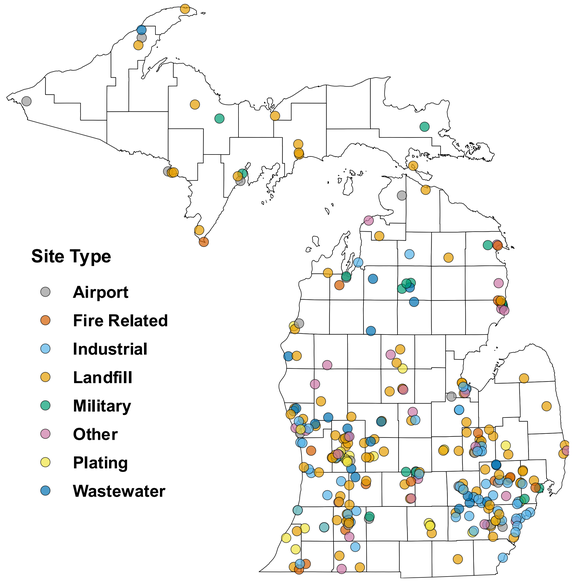


Hazard Index at Sample Sites





PFAS Contamination Sites



Conclusion

- PFHxS dominates as the analyte with the greatest impact on hazard index levels in drinking water samples.
- Many sites with the highest hazard index levels are school facilities.
- There are many drinking water sites and systems with hazard index greater than 1 which surpasses the set Maximum Contamination Level.
- We are also developing an interactive dashboard which will be published on GitHub: https://github.com/Nancy-Odhiambo/Michigan-PFAS-Data

References

1. Environmental Protection Agency. (2025, September 30). PFAS explained. https://www.epa.gov/pfas/pfas-explained
2. Michigan Department of Environment, Great Lakes, and Energy (EGLE). (2026). MPART PFAS Geographic Information System [Data set]. Michigan PFAS Action Response Team. https://www.michigan.gov/egle/maps-data/mpart-pfas-gis
3. U.S. Environmental Protection Agency. (2024, April 8). Understanding the final PFAS national primary drinking water regulation: Hazard index maximum contaminant level [Fact sheet]. https://www.epa.gov/system/files/documents/2024-04/pfas-npdwr_fact-sheet_hazard-index_4.8.24.pdf